

Getting the Most

from .NET MAUI

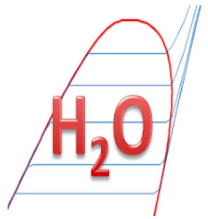
David Ortinau

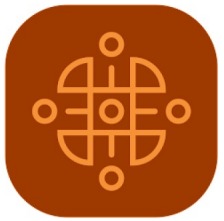
Principal Product Manager
.NET MAUI

@davidortinau






STOP!

H2O

aps






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Colonial Williamsburg

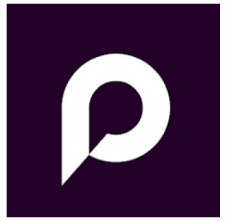
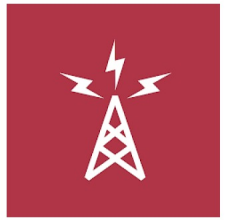

TRACKMAN

METRO
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SCOR
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QSRonline

SHOP
TAX
FREE


afp
modelo



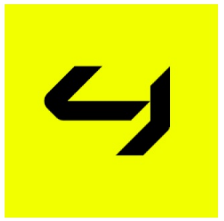
E-ZPass
TOLL BY PLATE

DigiD





we


MARCO'S
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TwinCAT
EchoGuide
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ECO TIPS
UZMAN TIPS
weyou app
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Mainland Football
BAJit
APA
GYS
COCO PLAY
BRIDGE
kamstrup
PRO PARKINGS
MAUI
DigiD
Christelijk Omroep
CONNECT
585 RETAIL
ONE PASS
TRACKMAN
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PEKA
limolink
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SKF
SSW
Skellefteå Kraft
FULL PULL
SCORES AND ODDS
COLLINS ROAD THEATRES
Ascend EVV
MOORE THEATRES
BYH
LINEUP HQ FANDUEL
Radio Message
REGENCY THEATRES
METRO ONE
Birthday Status
PSE
ST. MICHAEL CINEMA
QSRonline
My Vaccinations
RS
The British Horse Society
Create and Craft
Coyote
DB Client
ATHLETIC FITNESS
aps


2.5x
growth since .NET 8 release

.NET MAUI Customer Showcase



Know your goals

How much does your project need…



Max Code Share



Performance at all costs



Platform Consistency

Max Code Share

All .NET MAUI -> 95-100% share

Prefers platform behavior

Make use of

- Styles
- OnPlatform
- OnIdiom
- Adaptive Triggers
- Single project resources

The 5%

- Binding libraries
- Control customizations
- Custom app icons, splash
- Platform specifics

Performance

.NET for Android, .NET for iOS have hands-down the best perf.

Code share will be closer to 40-60%

Requires expertise in

- UIKit, Storyboards, Xcode
- Android widgets, XML layouts, Android Studio

Sharing

- All the business logic
- MAUI Embedded UI
- MAUI services

Not shared

- All the UI
- App icons, splash screens
- Platform behavior/services

Performance

Know your perf options

Trim Modes

- Partial
- Full

Android

- Profiled AOT
- Full AOT

iOS Native AOT

Native AOT is extremely aggressive and requires that ALL assemblies are trimmable

gist.github.com/ivanpovazan/4756bdb65a34366f0944e99f601379f3

github.com/jonathanpeppers/Mono.Profiler.Android

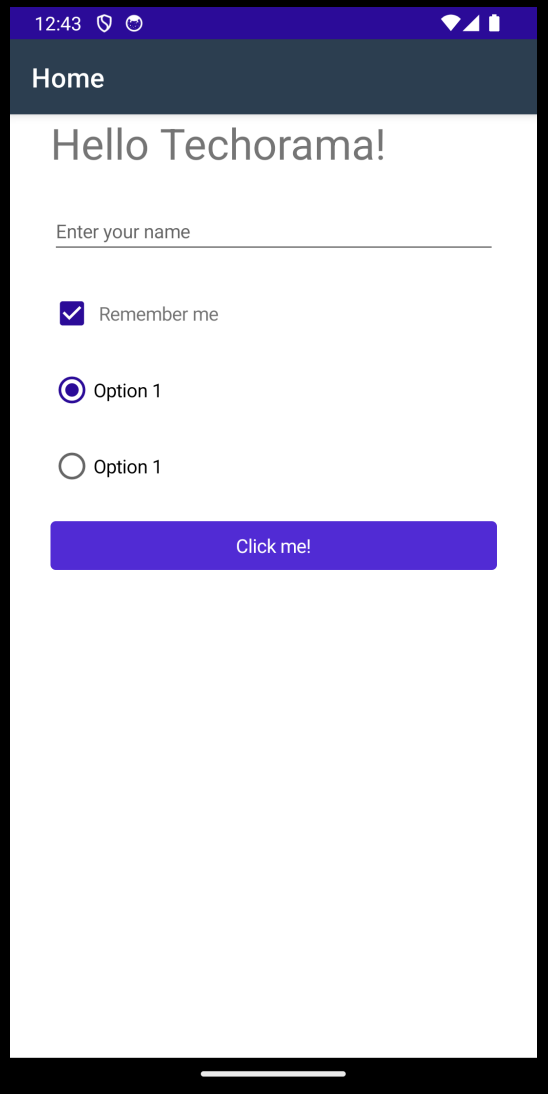
Platform Consistency

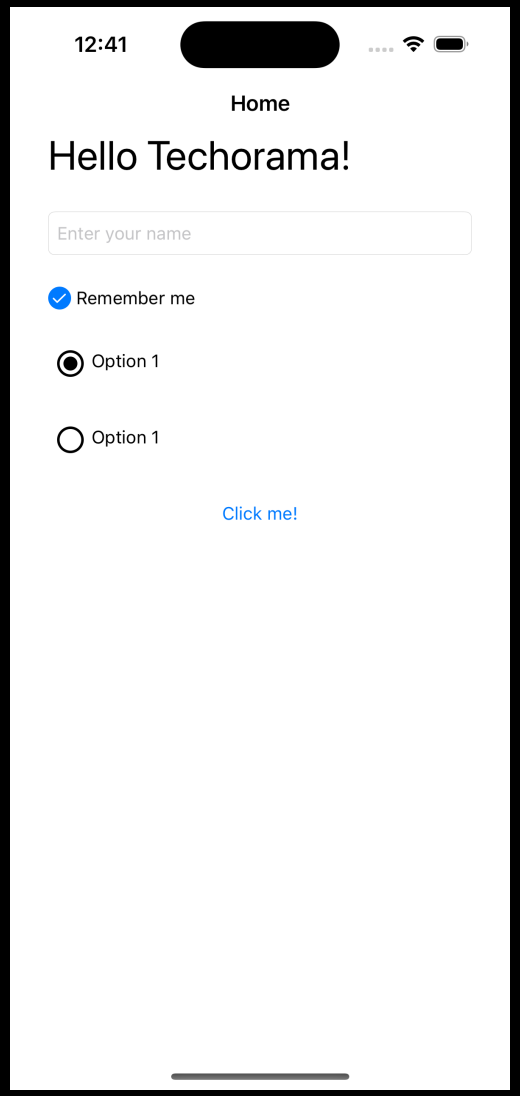
By default .NET MAUI preserves platform native behavior

Styling can get you close or the same on most platforms

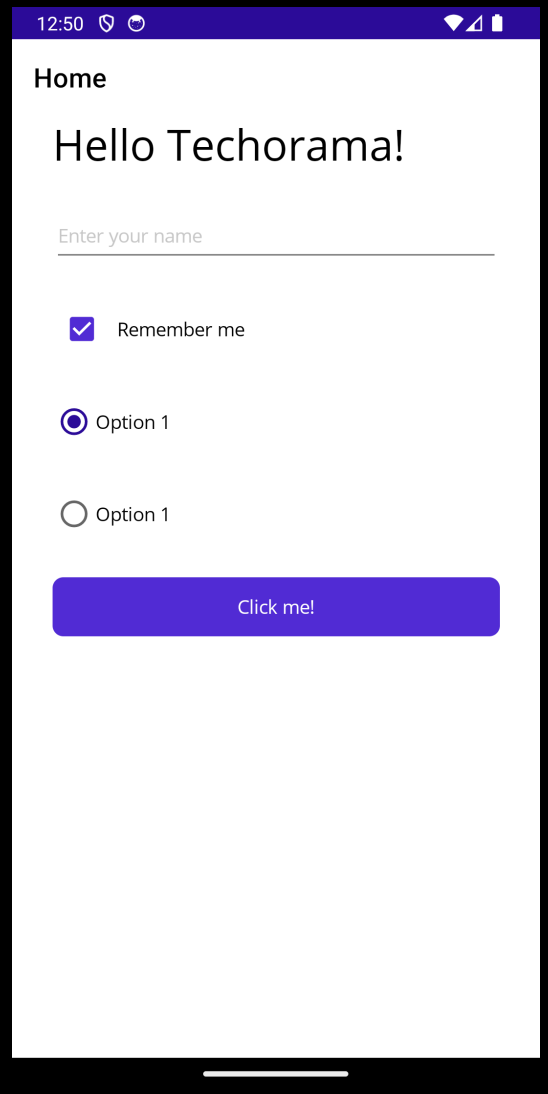
Maui.Graphics or Skia based controls are the best solution to ensure consistent look and feel anywhere

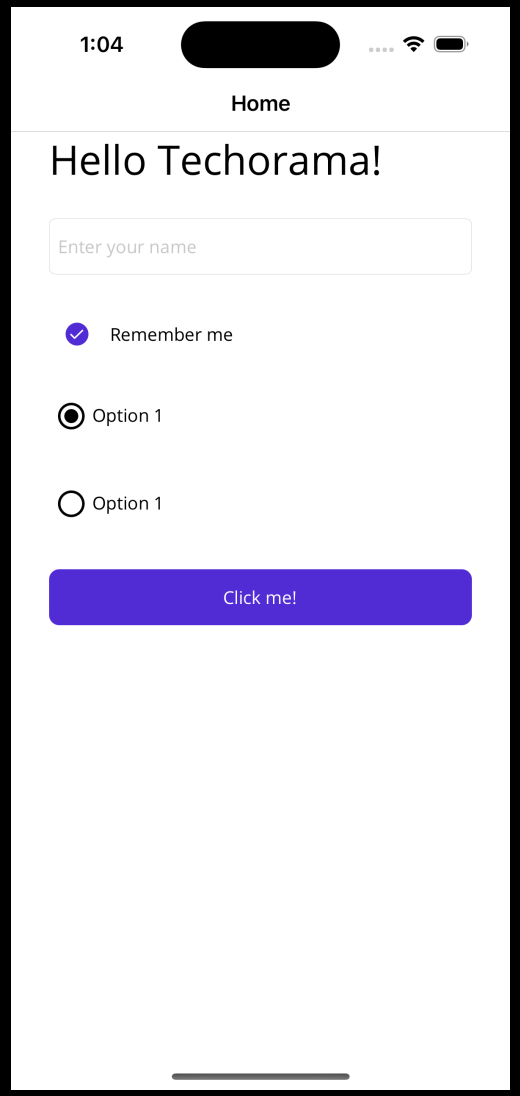
No Styling



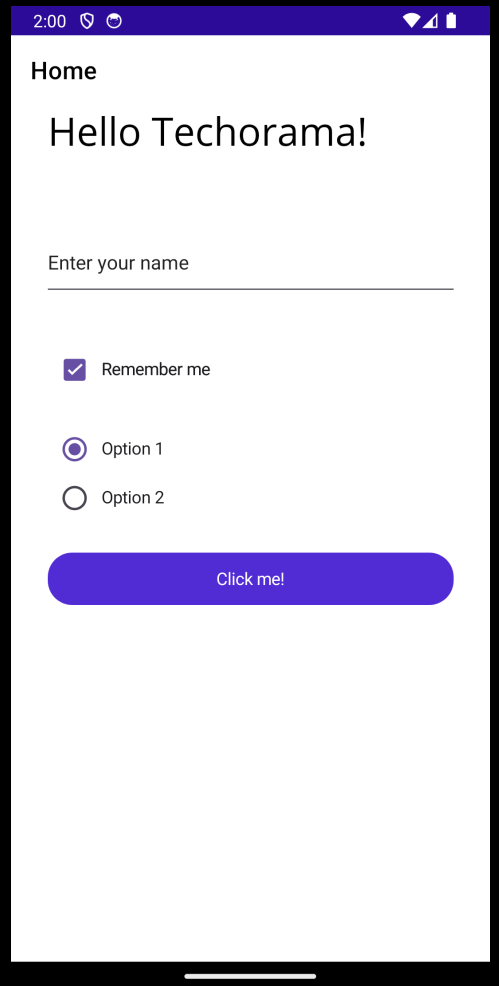


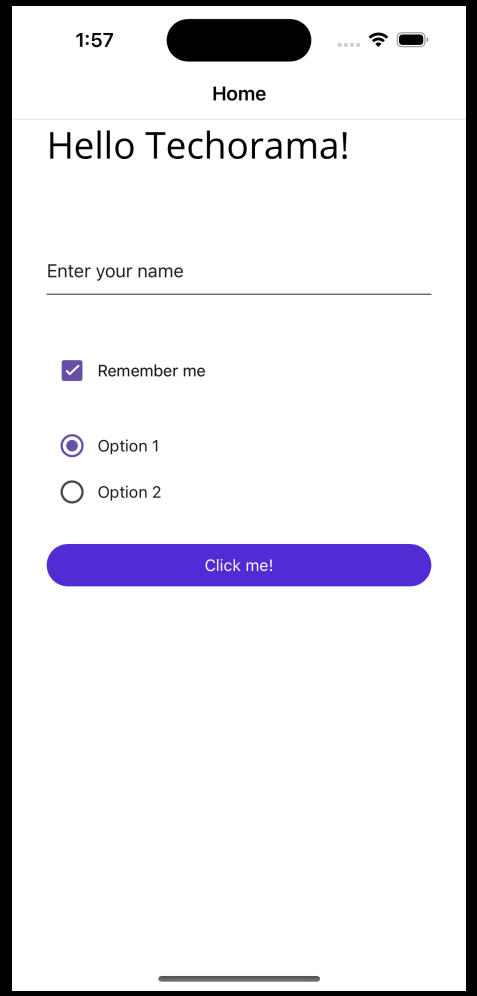
Basic Styling





Microsoft.Maui.Graphics





Know your options

Look at all the things you could use

In the box features and controls

Vendors

Single project

Multi-project

Font Icons

Svgs

Maui.Graphics

SkiaSharp

Native embedding

Hybrid

XAML

C# markup

Maui Reactor

MVVM vs MVU vs Reactive

NuGets

GitHub & Azure

Controls

		CarouselView	CollectionView	RefreshView
IndicatorView	Checkbox	RadioButton	ImageButton	SwipeView
ActivityIndicator	BoxView	Button	DatePicker	Editor
Entry	Image	Label	ListView	Map
OpenGLView	Picker	ProgressBar	SearchBar	Slider
Stepper	TableView	TimePicker	WebView	EntryCell
ImageCell	SwitchCell	TextCell	ViewCell	Path
Ellipse	Polygon	Rectangle	Line	Polyline

Native UI

Elements are realized via platform handlers.

	ContentPage	Label	Picker
Android	Activity / Fragment	TextView	AlertDialog + LinearLayout + EditText + NumberPicker
iOS/Mac	UIViewController	UILabel	UITextField + UIPickerView + UIToolbar + UIBarButtonItem
WinUI	Page	TextBlock	ComboBox

Component Vendors



















Know your tools

Look at all the things you could use

VS on Windows
- Live Preview
- Live Property
- Live Visual Tree

VS Code anywhere

Copilot

Rider

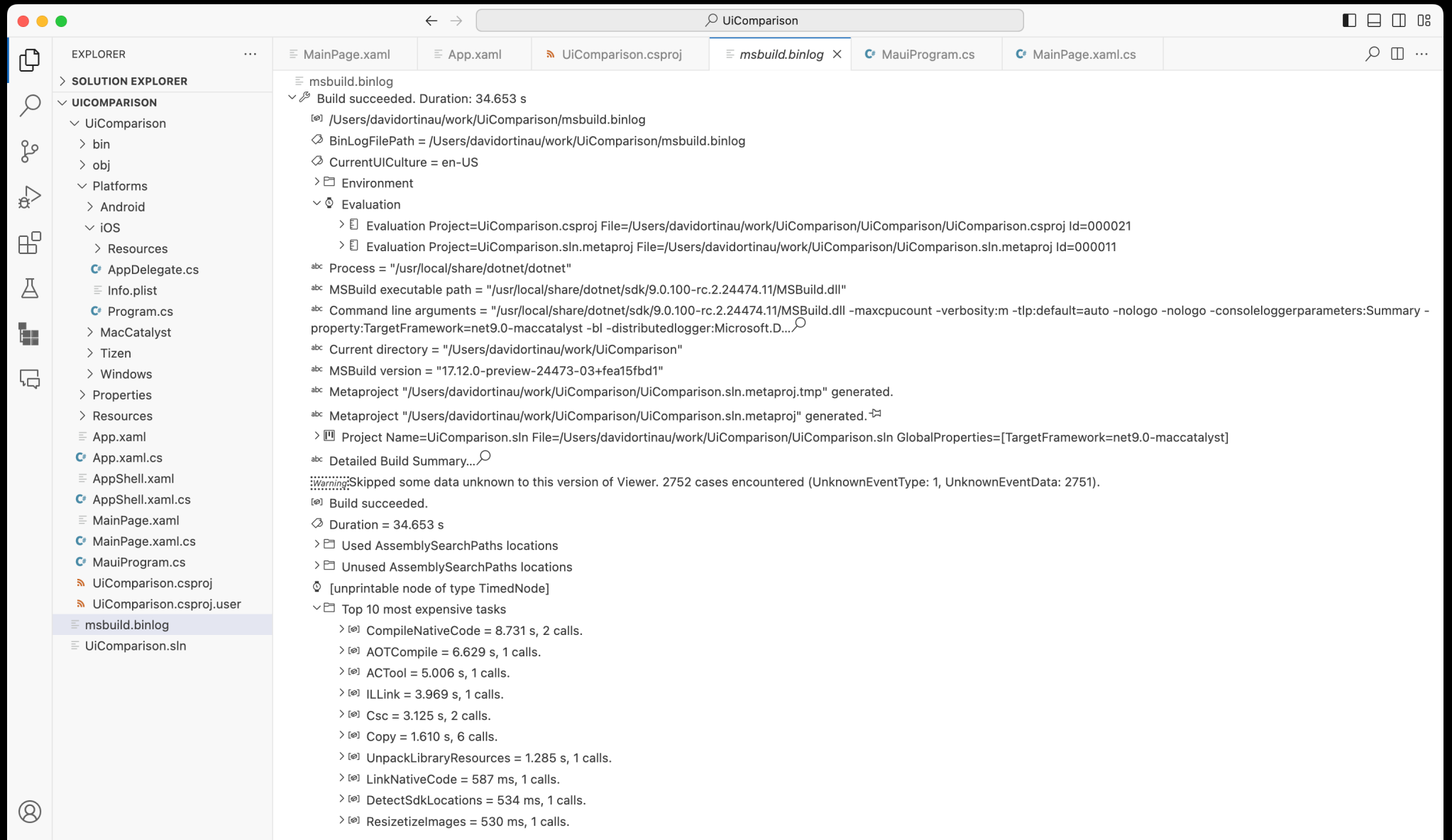
Speedscope

Binlogs

C# hot reload

XAML hot reload

dotnet build –f net9.0-android -bl



Know your hill to die on

Probably not

We don’t write ANY platform code

It must be THIS control

We ONLY use Stable releases

We only use LTS releases

We don’t use non-Microsoft dependencies

The app must match our design file EXACTLY

We must be able to test EVERYTHING

We will NOT fork anything ever

We will NOT contribute to the product

Know your support options

How to get support

Free and open source

- GitHub issues
- GitHub discussions
- VS Feedback

Paid

- Microsoft Unified Support

Discord – contributors

maui-upgrades@microsoft.com
for Xamarin upgrade questions

Thanks! Let’s talk.

@davidortinau

